

TECHNOLOGY CONSULTING

IN THE GLOBAL COMMUNITY

Final Consulting Report

Palau BoT

Preksha Goyal

August 2025

Carnegie Mellon University





BUREAU OF
TOURISM

Executive Summary

Preksha Goyal | Student Consultant, Carnegie Mellon University
Elangel Yanni Mesubed | Director, Bureau of Tourism, MHRCTD

I. About the Organization

The Bureau of Tourism (BoT) sits within Palau's Ministry of Human Resources, Culture, Tourism & Development and serves as the national steward of sustainable tourism. With a staff of just three—a Director and two Compliance Officers—the Bureau carries a mandate that spans licensing, policy formation, monitoring of visitor sites, and coordination with all sixteen state governments. Its guiding blueprint is the Palau Sustainable Tourism Strategy (PSTS) 2025-2028, which commits the nation to growth that protects reefs, heritage, and community livelihoods. BoT operates from the “One-Stop Shop” complex in the heart of Koror, an open and well-equipped workspace with high-speed connectivity and capacity for future hires. Day-to-day digital work happens on Dell workstations using Google Workspace and Microsoft Office, while core IT support is provided by a central government team. Until this engagement, most operational data—tour-operator licences, visitor-attraction status, and state project updates—lived in scattered spreadsheets or paper folders, forcing leadership to make decisions with months-old information and to piece together grant applications by hand.

II. Goal 1: Digitalise & Streamline State-Level Tourism Data Collection

To replace that fragmentation, the consulting team designed and launched the Destination Information Management System (DIMS). All state-level tourism assets and priorities were first mapped into a clean, relational schema in Google Sheets, then surfaced through a no-code AppSheet application that works online or offline and syncs automatically once a connection returns—crucial for out-island officials. Granular security rules let each user view and edit only the records that matter to them, while a built-in verification layer routes every proposed change through a “maker-checker” flow so data quality remains high. Dashboards now update in near real time, giving BoT an instant picture of fees, readiness, and capacity across all Palauan visitor attractions instead of waiting half a year for manual compilations. State governors, after live demonstrations, endorsed the platform enthusiastically and began entering updates within days; initial uptake shows over eighty percent of required fields populated inside the first fortnight. The result is a single source of truth that shrinks data-collection lead time from roughly 180 days to a matter of minutes and frees Bureau staff to focus on analysis rather than chasing spreadsheets.

III. Goal 2: Build Project-Management & Grant-Tracking Workflows

Building on this consulting project's foundation, DIMS now links every state to projects detailing measurable success criteria, subtasks, budgets, and target dates, while a parallel repository catalogs grants and training programs aligned to PSTS themes. When a state identifies a relevant opportunity, a simple "Show Interest" button triggers an automated workflow: the expression of interest is logged, the responsible Compliance Officer is notified, and the state's project record is updated to reflect pending funding. Because each grant is tagged to strategic priorities, BoT can instantly see where financing gaps remain and where donor resources might achieve the greatest impact. The integrated view has already replaced sixteen separate email chains with one coordinated pipeline, cutting outreach effort by more than ninety percent and ensuring that all projects carry uniform reporting standards. Looking ahead, quarterly briefings generated directly from DIMS will give donors transparent evidence of progress, while performance-linked disbursements and light-touch incentives are set to deepen state engagement. Together, the data-collection and project-management tracks transform BoT from a paper-bound regulator into a digitally empowered orchestrator of sustainable growth, positioning Palau as a regional exemplar of how lean institutions can harness no-code technology for high-value public outcomes.

Consulting Partner

Elangel Yanni Mesubed
eymesubed@gmail.com

Ministry of Human Resources, Culture,
Developments and Tourism
One Stop Shop, Koror, Palau
<https://ministryofhrctd.pw/>

About the Consultant

Preksha Goyal
prekshag@andrew.cmu.edu

Preksha is a Master Student in ETIM.
She's going back to Pittsburgh for her next semester
with a goal of working as a product manager.



BUREAU OF TOURISM

Final Consulting Report

Preksha Goyal | Student Consultant, Carnegie Mellon University
Elangel Yanni Mesubed | Director, Bureau of Tourism, MHRCTD

I. About the Organization

Organization

The mission of the Bureau of Tourism (BoT) is to support sustainable tourism development that benefits both the environment and local communities, in alignment with Palau's conservation and cultural values.

The Bureau of Tourism (BoT), housed within the Ministry of Human Resources, Culture, Tourism & Development (MHRCTD) of the Republic of Palau, exists to cultivate a tourism sector that sustains economic growth *and* safeguards Palau's unique environment and culture. Its primary policy vehicle is the **Palau Sustainable Tourism Strategy (PSTS) 2025-2028**, which codifies national priorities for responsible visitation, ecological preservation, and data-driven governance.

Facilities

- Modern building called "One Stop Shop" with various ministry offices inside.
- Located in the centre of Koror city on Main Street
- The MHRCTD has its own office inside the building with different sections allotted to 4 different bureaus of the ministry, one section allotted to BoT.
- Office is an open and airy space; ample empty space for future hires.
- Each employee has a wide desk with drawers/shelves, DELL monitor, bluetooth peripherals and office extension landlines.
- The office is well lit and air conditioned, with high speed WiFi, extensions to plug devices

Programs

MHRCTD provides services of licensing, standards, monitoring, inter-agency coordination across 16 states, and donor engagement.

Staff

Headcount: 1 Director, 2 Compliance Officers (with plans to double by 2026)

- Elangel Yanni Mesubed | Director
 - Drives Palau Tourism Strategy

- Oversees all BoT operations
- Responsible for securing funding
- Endira Absalom | Compliance Office
 - Registers, licenses and grades tourism businesses
 - Handles back office operation
- Indirah Tirso | Compliance Officer
 - Registers, licenses and grades tourism businesses
 - Handles front office operations

The team is adaptable and highly motivated to digitize their operations.

Everyday Applications: Microsoft Office, Gmail, GDrive, Google Chrome

Technology Infrastructure

Hardware	DELL monitors, bluetooth mouse and keyboards
Installed Applications	Microsoft Office, Whatsapp, Chrome
OS	Windows
Wi Fi	High Speed; 150-200 MBPS on average

Technology Management

- An IT team hired by the Ministry of Finance is responsible for installing OS, maintaining technology infrastructure and fixing problems on an ad-hoc basis, for all government offices.
- A consulting IT team in Indonesia builds websites and digitizes various processes for all government offices.
- A permanent IT lead has been hired, they'll start working with BoT from 28 July, 2025

Technology Planning

- Project and technology planning is led by the Director and currently implemented by consultants.
- The full-time IT lead will focus on technology planning once they join.
- The PSTS strategy includes clear project-related goals but leaves technological implementation details to consultants

Communication

- Channels: Email, WhatsApp, phone
- Physical applications for tour operators; manually recorded in Google Sheets
- Lack of organised communication/collaboration between state governments in MHRCTD

Information Management

Important Information	Tour Operators Application & License Information Boats Application & License Information Rules & Regulations for tourism bodies
Management	Mixed (Paper, Digital)
IT Work	IT Consultants, Interns

Business Systems

Finances including payrolls, budget approvals etc are handled by offices in Ministry of Finance

II. Goal 1: Digitalise & Streamline State-Level Tourism Data Collection

Motivation

Before the consulting project started, BoT collected state data via sporadic emails, calls, and paper forms, resulting in at least lags and policy decisions made with stale or inaccurate information. States need an intuitive way to view goals, report progress, and update data—even in low-connectivity areas. They expressed concern with the MHRCTD about the lack of communication between the ministry and state governments as well as lack of collaborative action.

Outcomes

Activities Delivered:

- Conducted user-journey interviews and demos with state governors and/or tourism officials across all states.
- Explored and evaluated the feasibility of a range of potential technology platforms to support the Bureau's tourism data management goals
- Extracted features from unorganised state-level tourism data and organized them into a structured schema, populated the data in Google Sheets
- Built a low-code front-end in AppSheet featuring offline sync, two-way data binding, granulated user access.
- Implemented live dashboards for tracking tourism assets including visitor attractions, future opportunities and upcoming state-level tourism priorities.
- Implemented an announcement mechanism to send push notifications in the form of emails to multiple users in a one go
- Trained BoT staff and state focal points; created user manuals for users as well as admin to continue developing the platform post internship.
- Verification Layer was introduced to prevent states from directly editing information - instead go through a maker-checker mechanism with BoT admin as checker

Key Outputs:

- Subjective data organised into schemas in Google Sheets
- DIMS v1 is live and deployed. States governors expressed great excitement and acceptance of the platform.
- Mobile-friendly app installable on any Android/iOS device

Key Outcomes/Indicators:

- Central digitised application for data collection reduced data reporting time from over 6 months to almost instantaneous
- Fragmented information from 16 states and MHRCTD, compiled into a single application

Configurations:

The Destination Information Management System (DIMS) was built entirely using AppSheet, a no-code platform that integrates with Google Sheets and provides both web app and mobile app(iOS, Android). The configuration phase included setting up over **14 interrelated tables**, each corresponding to a core module such as *visitor attractions*, *state priorities*, *projects*, *grants*, and *user permissions*. Critical configurations also included:

- **Offline mobile compatibility** for remote state access
- **Security, View and Access configurations** ensuring users could only view and edit content relevant to them
- **Automated email workflows** triggered by various actions to notify BoT/states.
- **UX customizations**, such as conditional views and format rules, to improve usability
- **Inline dashboards** for real-time visualization of project progress and grant availability

Recommendations

1. Nominate a DIMS Coordinator at BoT

Why this matters:

Currently, system updates, user access control, and content follow-up are ad hoc. Without a designated internal owner, there's a risk of DIMS stagnating or becoming outdated. A DIMS Coordinator would oversee onboarding of new state users, respond to support issues, and lead monthly data validation checks. This ensures continuity, reduces consultant dependence, and sustains impact.

Impact & ROI:

Assigning a staff member for just 4–6 hours per week would preserve the ~85% time savings already achieved on data collection, reducing reliance on manual follow-ups. It would also support continuous adoption across all 16 states.

Implementation Steps:

- Appoint a BoT staff member to act as DIMS point-of-contact
- Provide them 1-on-1 AppSheet training using existing user manual & AppSheet documentation
- Define their monthly tasks: onboarding, usage tracking, data integrity checks

- Set quarterly DIMS review meetings for oversight and feedback

Resources Required:

- Consultant-created user manual
- AppSheet’s own documentation and forums for troubleshooting
- Internal BoT calendar to set recurring “data sprint” checkpoints

2. Integrate Usage Analytics to Trigger Coaching

Why this matters:

While the system is in place, usage varies widely by state. Some states are updating regularly, while others have remained inactive. By tracking user activity (logins, submissions, last update timestamps), BoT can identify where additional support or nudging is needed.

Impact & ROI:

Early detection of disengaged states allows for timely outreach. This avoids the data collection lag BoT previously experienced and maintains data completeness at over 90%, boosting accuracy for grant funding and planning.

Implementation Steps:

- Create a “Last Updated” virtual column in key tables (e.g., state profiles, projects)
- Add a summary dashboard view for BoT showing which states are active/inactive
- Assign the DIMS Coordinator to follow up with low-engagement states via call or visit
- Incorporate usage review into monthly staff meetings

Resources Required:

- AppSheet expressions for calculating last edit timestamps
- Built-in AppSheet dashboard for quick visual tracking
- BoT team calendar to trigger follow-ups

3. Grant Read-Only Access to Ministry of Finance and Palau Visitors Authority

Why this matters:

DIMS currently empowers BoT and states, but not key stakeholders like the Ministry of Finance or Palau Visitors Authority. Granting them view-only access would enhance national coordination and reduce delays in approvals, budget allocation, and reporting.

Impact & ROI:

Reducing email-based status updates could save BoT staff ~5 hours/month in reporting time. It would also increase credibility with funding agencies by demonstrating a real-time, transparent planning tool.

Implementation Steps:

- Add selected users from the Ministry and PVA with view-only roles in AppSheet

- Use existing AppSheet security filters to restrict editing
- Walk these stakeholders through the dashboard and use cases
- Offer quarterly briefings that align with budget or tourism planning cycles

Resources Required:

- A short “viewer guide” to help new users navigate
- Support from the DIMS Coordinator to respond to access or navigation issues

4. Periodically Backup All DIMS Data

Why this matters:

DIMS runs on Google Sheets/AppSheet, which is reliable but still susceptible to data overwriting or accidental deletions. Regular backups ensure the integrity of this mission-critical data and protect institutional memory.

Impact & ROI:

Avoiding a single data loss event would save weeks of recovery and reduce the risk of delayed licensing, grants, or audits. It also strengthens BoT’s data governance credentials.

Implementation Steps:

- Use Google Sheets’ “Version History” feature weekly to save snapshots
- Alternatively, duplicate the primary sheet once a month/quarter into a “Backup” folder
- Label backups by date and automate this task using Google Workspace tools or calendar reminders
- Store copies in a secure Google Drive folder with restricted access

Resources Required:

- Google Workspace (already in use)
- Internal calendar and reminders for monthly backup tasks
- Optional: Google Apps Script to automate snapshotting

III. Goal 2: Build Project-Management & Grant-Tracking Workflows

Motivation

BoT lacked visibility into which projects states were pursuing, states lacked visibility in which grants were available, and how those pieces connected to PSTS priorities.

Outcomes

Activities Delivered:

- Developed a **project management module** (Project, Tasks, Performance Criteria, Date-Wise Targets, Comments).

- Curated a **grant/training program repository** with the ability of linking programs to state priorities via a pre-defined set of priority categories.
- Created an “**Show Interest**” flow—submitting a grant EOI triggers an AppSheet bot, logging the request and alerting BoT.
- Embedded budget management in the projects to allow for digitised reporting and supporting information to apply to grants.

Key Outputs:

- BoT can now **cross-reference** grant availability with state priorities, accelerating resource matching and donor reporting
- 10 new interconnected tables with multiple sub-schemas added to the application
- Show Interest Action to allow users to express interest to MHRCTD in trainings/grants
- Project management functionality with complete ownership to states to add/update/delete projects

Key Outcomes/Indicators:

- Centralized grant visibility reduces effort duplication across states. 93% reduction in outreach load (1 coordinated source vs. 16 separate efforts)
- Project tracking system formalizes implementation monitoring across all states. 100% of projects include success criteria and subtasks - making reporting across states uniform
- Grants are now categorized by priority types. 100% of grant records tagged with one or more thematic categories

Configurations:

The **Grants Repository** and **Project Management** modules were designed and configured in AppSheet to support BoT’s goals of transparent funding, better alignment with state needs, and structured tracking of tourism initiatives.:

- Grants were stored in a Google Sheet with key attributes such as title, description, funding agency, and category (linked via **EnumList** to development priorities).
- An **expression-of-interest (EOI)** form was built using AppSheet’s form view. When submitted, it triggers a custom **email notification workflow** to alert BoT
- **Virtual columns were configured to calculate progress, allowing real-time visualization of implementation.**

Recommendations

1. Issue Quarterly Progress Briefs from DIMS to Donors and Stakeholders

Why:

BoT can increase transparency and donor confidence by turning real-time data into digestible quarterly updates. This reduces manual report preparation and demonstrates the impact of grants and state activities.

Impact:

BoT staff currently spend hours compiling updates manually; auto-generated briefs could reduce reporting prep time by 75%. They also strengthen BoT's credibility and funding prospects.

How:

- Use existing dashboards in AppSheet to summarize state activity and grant interest
- Export key stats to a standardized briefing template each quarter
- Email to donor and inter-ministerial stakeholders

Resources:

- AppSheet's export capabilities + a Google Docs briefing template
- Support from the DIMS Coordinator to gather and format metrics

2. Link Future Grant Disbursement to Success Criteria Completion**Why:**

Linking disbursement to real progress (e.g., completed subtasks, updated priorities) reinforces accountability and ensures grants are used effectively.

Impact:

Aligns funding with measurable outputs, improving ROI and reducing dormant or misallocated grants. Can drive 20–30% higher grant engagement.

How:

- Add a grant eligibility dashboard that cross-references project progress
- BoT verifies completion via DIMS before approving next grant installment
- Communicate criteria clearly to state users during rollout

Resources:

- Existing project tracking module in DIMS
- Admin dashboard filters and virtual columns
- Basic policy document explaining grant-release conditions

3. Pilot KPI-Based Incentives for High-Performing States**Why:**

Offering benefits like fast-track licensing or bonus training slots to states with high DIMS participation will drive usage and healthy competition.

Impact:

Motivates lagging states, improving adoption and data quality. Even a small incentive could increase usage rates across states.

How:

- Define KPIs (e.g., % of fields updated, timely grant interest, dashboard views)
- Monitor engagement quarterly using AppSheet usage logs
- Offer soft incentives like recognition, priority support, or policy perks

Resources:

- Usage tracking via AppSheet analytics and logs
- Communication plan for rollout and recognition
- BoT leadership support to authorize incentives

▪

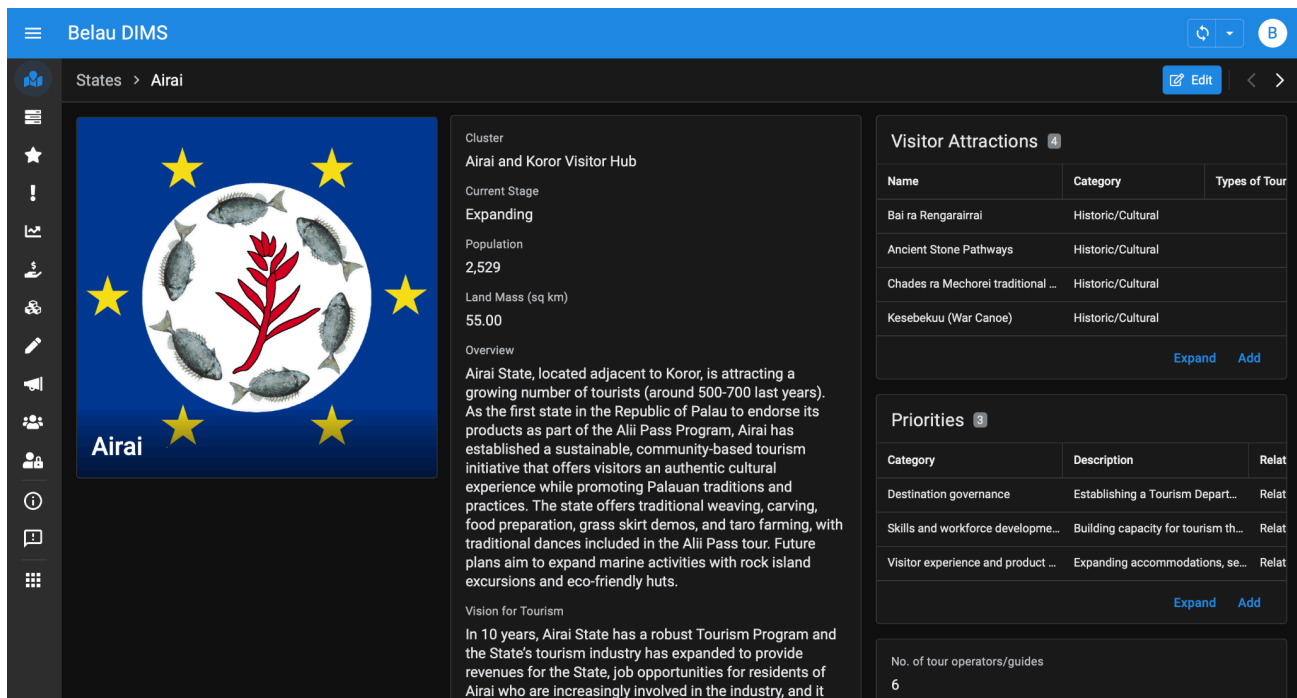
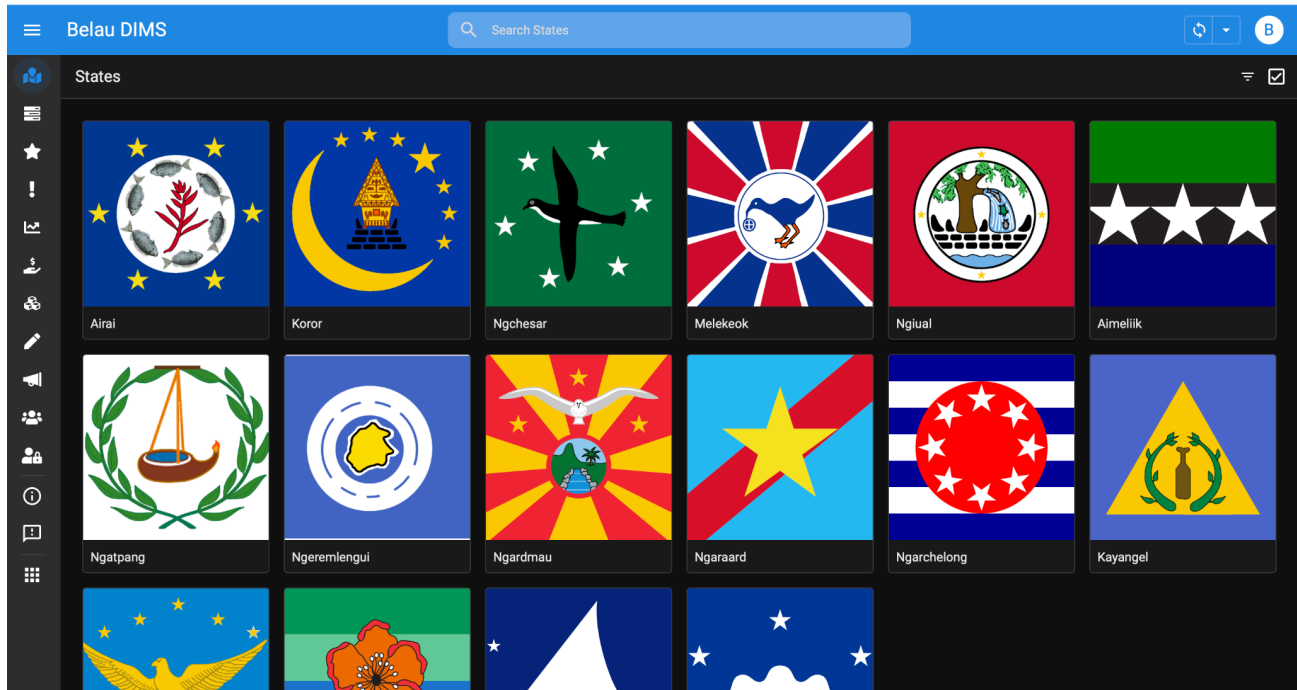
About the Consultant

Preksha Goyal is a candidate for the M.S. in Engineering & Technology Innovation Management at Carnegie Mellon University and a former software engineer at Goldman Sachs. Specialising in data platforms and human-centred digital strategy, she partnered with the Bureau of Tourism from May to July 2025 as an MBB-style consultant, guiding the end-to-end design and launch of DIMS. Her work leaves BoT with a scalable, locally owned system that advances the PSTS, elevates evidence-based decision-making, and positions Palau as a regional exemplar of sustainable tourism governance.

Appendix A. Project Questions, Considerations, platform comparisons

[Project Considerations](#)

Appendix B. App Screenshots



Belau DIMS										
Search Projects										
Projects										
Project	Entity	Status	Created By	Assigned To	Anticipated Start Date	Anticipated End Date	Actual Start Date	Actual End date	Estimated Cost	
Project 2	Airai	Not Started	BoT	YWND	26/6/2025	31/7/2025	26/6/2025	26/6/2025	\$50,000	
Project	Kayangel	In Progress	Preksha	Preksha	25/7/2025	16/8/2025	24/7/2025		\$5,000	

<

</

Belau DIMS								
Available Programs								
Program Title	Program Agency	Description	Type	Category	Available Amount(...	Application Deadline	Program Sta	
Tourism Marketing and Promoti...	Japan International Cooperatio...	This program is primarily desig...	Training	Skills and workforce developme...	0	4/9/2025 11:59:59 PM		
Human Capital Development fo...	Japan International Cooperatio...	focusing on training individuals ...	Training	Visitor experience and product ...				
Sustainable Tourism Resources...	Japan International Cooperatio...		Grant	Destination management , Desti...				
Planning and Management of N...	Japan International Cooperatio...		Training	Destination management , Desti...				
Museums and Community Deve...	Japan International Cooperatio...		Training	Skills and workforce developme...				
ICT Core Personnel Developme...	Japan International Cooperatio...		Training	Skills and workforce developme...				
Maritime Search and Rescue, Di...	Japan International Cooperatio...		Training	Destination management , Sust...				
Countermeasures Against Cybe...	Japan International Cooperatio...		Training	Destination management , Desti...				
Power Grid Planning and Operat...	Japan International Cooperatio...		Training	Destination management , Desti...				
Coastal and Marine Ecosystem ...	Japan International Cooperatio...		Training	Destination management , Desti...				
Non-Revenue Water Manageme...	Japan International Cooperatio...		Training	Destination management , Desti...				
Tourism Marketing and Promoti...	Japan International Cooperatio...		Training	Destination management , Desti...				
U.S. Dept. of the Interior - OIA ...								
Republic of Palau Formula Educ...								
Micronesia Conservation Trust ...								

